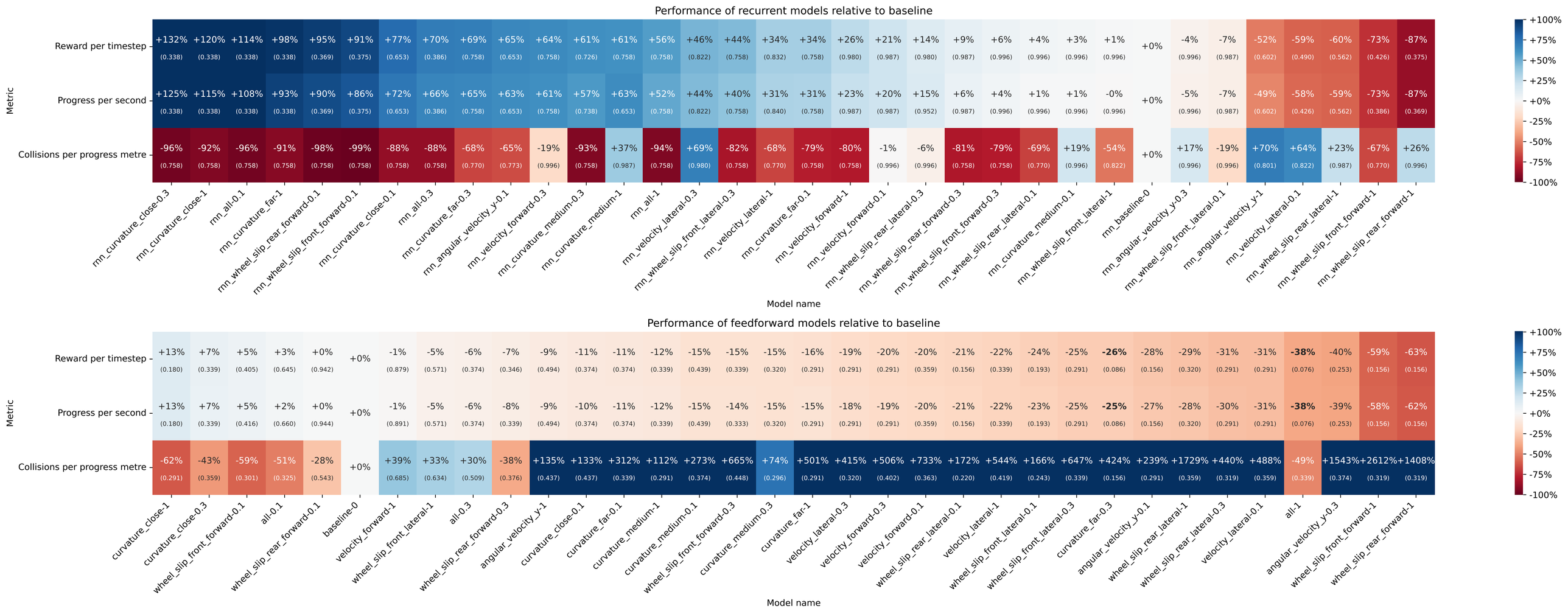
Performance of recurrent models relative to baseline
Metric
Reward per timestep
Progress per second
Collisions per progress metre
+132% (0.338) +120% (0.338) +114% (0.338) +98% (0.338) +95% (0.369) +91% (0.375) +77% (0.653) +70% (0.386) +69% (0.758) +65% (0.653) +64% (0.758) +61% (0.726) +61% (0.758) +56% (0.758) +46% (0.822) +44% (0.758) +34% (0.832) +34% (0.758) +26% (0.980) +21% (0.987) +14% (0.980) +9% (0.987) +6% (0.996) +4% (0.996) +3% (0.996) +1% (0.996) +0% -4% (0.996) -7% (0.987) -52% (0.602) -59% (0.490) -60% (0.562) -73% (0.426) -87% (0.375)
+125% (0.338) +115% (0.338) +108% (0.338) +93% (0.338) +90% (0.369) +86% (0.375) +72% (0.653) +66% (0.386) +65% (0.758) +63% (0.653) +61% (0.758) +57% (0.738) +63% (0.653) +52% (0.758) +44% (0.822) +40% (0.758) +31% (0.840) +31% (0.758) +23% (0.987) +20% (0.987) +15% (0.952) +6% (0.987) +4% (0.996) +1% (0.996) +1% (0.996) -0% (0.996) +0% -5% (0.996) -7% (0.987) -49% (0.602) -58% (0.426) -59% (0.562) -73% (0.386) -87% (0.369)
-96% (0.758) -92% (0.758) -96% (0.758) -91% (0.758) -98% (0.758) -99% (0.758) -88% (0.758) -88% (0.758) -68% (0.770) -65% (0.773) -19% (0.996) -93% (0.758) +37% (0.987) -94% (0.758) +69% (0.980) -82% (0.758) -68% (0.770) -79% (0.758) -80% (0.758) -1% (0.996) -6% (0.996) -81% (0.758) -79% (0.758) -69% (0.770) +19% (0.996) -54% (0.822) +0% +17% (0.996) -19% (0.996) +70% (0.801) +64% (0.822) +23% (0.987) -67% (0.770) +26% (0.996)
rnn_curvature_close-0.3
rnn_curvature_close-1
rnn_all-0.1
rnn_curvature_far-1
rnn_wheel_slip_rear_forward-0.1
rnn_wheel_slip_front_forward-0.1
rnn_curvature_close-0.1
rnn_all-0.3
rnn_curvature_far-0.3
rnn_angular_velocity_y-0.1
rnn_velocity_forward-0.3
rnn_curvature_medium-0.3
rnn_curvature_medium-1
rnn_all-1
rnn_velocity_lateral-0.3
rnn_wheel_slip_front_lateral-0.3
rnn_velocity_lateral-1
rnn_curvature_far-0.1
rnn_velocity_forward-1
rnn_velocity_forward-0.1
rnn_wheel_slip_rear_lateral-0.3
rnn_wheel_slip_rear_forward-0.3
rnn_wheel_slip_front_forward-0.3
rnn_wheel_slip_rear_lateral-0.1
rnn_curvature_medium-0.1
rnn_wheel_slip_front_lateral-1
rnn_baseline-0
rnn_angular_velocity_y-0.3
rnn_wheel_slip_front_lateral-0.1
rnn_angular_velocity_y-1
rnn_velocity_lateral-0.1
rnn_wheel_slip_rear_lateral-1
rnn_wheel_slip_front_forward-1
rnn_wheel_slip_rear_forward-1
Model name
+100%
+75%
+50%
+25%
+0%
-25%
-50%
-75%
-100%
Performance of feedforward models relative to baseline
Metric
Reward per timestep
Progress per second
Collisions per progress metre
+13% (0.180) +7% (0.339) +5% (0.405) +3% (0.645) +0% (0.942) +0% -1% (0.879) -5% (0.571) -6% (0.374) -7% (0.346) -9% (0.494) -11% (0.374) -11% (0.374) -12% (0.339) -15% (0.439) -15% (0.339) -15% (0.320) -16% (0.291) -19% (0.291) -20% (0.291) -20% (0.359) -21% (0.156) -22% (0.339) -24% (0.193) -25% (0.291) **-26%** (0.086) -28% (0.156) -29% (0.320) -31% (0.291) -31% (0.291) **-38%** (0.076) -40% (0.253) -59% (0.156) -63% (0.156)
+13% (0.180) +7% (0.339) +5% (0.416) +2% (0.660) +0% (0.944) +0% -1% (0.891) -5% (0.571) -6% (0.374) -8% (0.339) -9% (0.494) -10% (0.374) -11% (0.374) -12% (0.339) -15% (0.439) -14% (0.333) -15% (0.320) -15% (0.291) -18% (0.291) -19% (0.291) -20% (0.359) -21% (0.156) -22% (0.339) -23% (0.193) -25% (0.291) **-25%** (0.086) -27% (0.156) -28% (0.320) -30% (0.291) -31% (0.291) **-38%** (0.076) -39% (0.253) -58% (0.156) -62% (0.156)
-62% (0.291) -43% (0.359) -59% (0.301) -51% (0.325) -28% (0.543) +0% +39% (0.685) +33% (0.634) +30% (0.509) -38% (0.376) +135% (0.437) +133% (0.437) +312% (0.339) +112% (0.291) +273% (0.374) +665% (0.448) +74% (0.296) +501% (0.291) +415% (0.320) +506% (0.402) +733% (0.363) +172% (0.220) +544% (0.419) +166% (0.243) +647% (0.339) +424% (0.156) +239% (0.291) +1729% (0.359) +440% (0.319) +488% (0.359) -49% (0.339) +1543% (0.374) +2612% (0.319) +1408% (0.319)
curvature_close-1
curvature_close-0.3
wheel_slip_front_forward-0.1
all-0.1
wheel_slip_rear_forward-0.1
baseline-0
velocity_forward-1
wheel_slip_front_lateral-1
all-0.3
wheel_slip_rear_forward-0.3
angular_velocity_y-1
curvature_close-0.1
curvature_far-0.1
curvature_medium-1
curvature_medium-0.1
wheel_slip_front_forward-0.3
curvature_medium-0.3
curvature_far-1
velocity_lateral-0.3
velocity_forward-0.3
velocity_forward-0.1
wheel_slip_rear_lateral-0.1
velocity_lateral-1
wheel_slip_front_lateral-0.1
wheel_slip_front_lateral-0.3
curvature_far-0.3
angular_velocity_y-0.1
wheel_slip_rear_lateral-1
wheel_slip_rear_lateral-0.3
velocity_lateral-0.1
all-1
angular_velocity_y-0.3
wheel_slip_front_forward-1
wheel_slip_rear_forward-1
Model name
+100%
+75%
+50%
+25%
+0%
-25%
-50%
-75%
-100%